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ADDITIONAL MATHEMATICS **4037/23**

Paper 2 **May/June 2010**

2 hours

Additional Materials: Answer Booklet/Paper
Graph paper (2 sheets)

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Mathematical Formulae

1. ALGEBRA

Quadratic Equation

For the equation $ax^2 + bx + c = 0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

Binomial Theorem

$$(a + b)^n = a^n + \binom{n}{1} a^{n-1} b + \binom{n}{2} a^{n-2} b^2 + \ldots + \binom{n}{r} a^{n-r} b^r + \ldots + b^n,$$

where n is a positive integer and $\binom{n}{r} = \frac{n!}{(n-r)!r!}$.

2. TRIGONOMETRY

Identities

$$\sin^2 A + \cos^2 A = 1.$$

$$\sec^2 A = 1 + \tan^2 A.$$

$$\operatorname{cosec}^2 A = 1 + \cot^2 A.$$

Formulae for ΔABC

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.$$

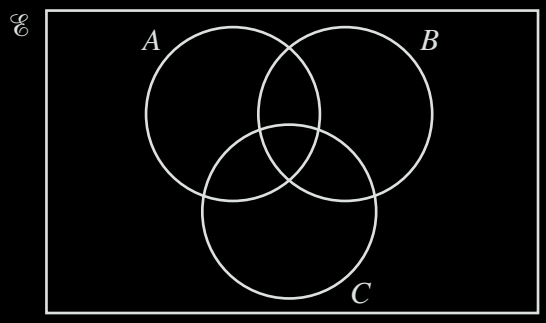
$$a^2 = b^2 + c^2 - 2bc \cos A.$$

$$\Delta = \frac{1}{2} bc \sin A.$$

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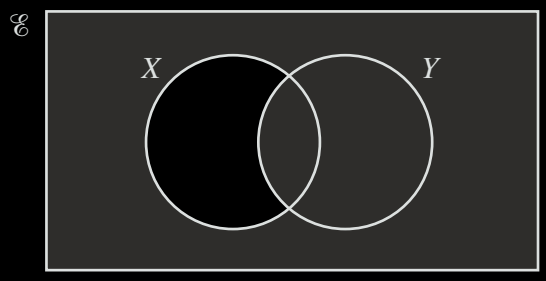
1 Find $\int\left(2 + 5x - \frac{1}{(x-2)^2}\right)\mathrm{d}x$.

2 **(a)**



Copy the diagram and shade the region which represents the set $A \cup (B \cap C')$. [1]

(b)



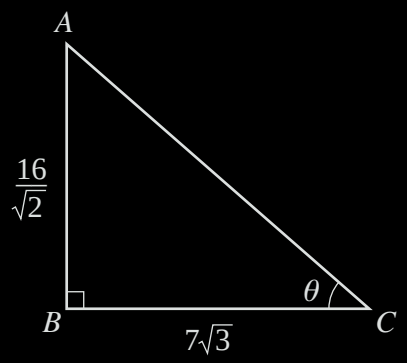
Express, in set notation, the set represented by the shaded region. [1]

(c) The universal set $\mathscr{E}$ and the sets P and Q are such that $\mathrm{n}(\mathscr{E}) = 30$, $\mathrm{n}(P) = 18$ and $\mathrm{n}(Q) = 16$. Given that $\mathrm{n}(P \cup Q)' = 2$, find $\mathrm{n}(P \cap Q)$. [2]

3 The volume $V\,\mathrm{cm}^3$ of a spherical ball of radius $r\,\mathrm{cm}$ is given by $V = \frac{4}{3}\pi r^3$. Given that the radius is increasing at a constant rate of $\frac{1}{\pi}\,\mathrm{cm\,s^{-1}}$, find the rate at which the volume is increasing when $V = 288\pi$. [4]

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4



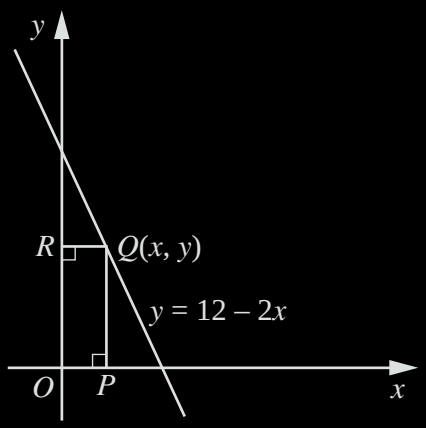
The diagram shows a right-angled triangle ABC in which the length of AB is $\frac{16}{\sqrt{2}}$, the length of BC is $7\sqrt{3}$ and angle BCA is θ.

(i) Find $\tan\theta$ in the form $\frac{a\sqrt{6}}{b}$, where a and b are integers. [2]

(ii) Calculate the length of AC, giving your answer in the form $c\sqrt{d}$, where c and d are integers and d is as small as possible. [3]

5 Solve the equation $2x^3 - 3x^2 - 11x + 6 = 0$. [6]

6



The diagram shows part of the line $y = 12 - 2x$. The point $Q(x, y)$ lies on this line and the points P and R lie on the coordinate axes such that $OPQR$ is a rectangle.

(i) Write down an expression, in terms of x, for the area A of the rectangle $OPQR$. [2]

(ii) Given that x can vary, find the value of x for which A has a stationary value. [3]

(iii) Find this stationary value of A and determine its nature. [2]

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7 **(i)** Sketch the graph of $y = |3x + 9|$ for $-5 < x < 2$, showing the coordinates of the points where the graph meets the axes.

(ii) On the same diagram, sketch the graph of $y = x + 6$.

(iii) Solve the equation $|3x + 9| = x + 6$. [3]

8 **(a)** **(i)** Write down the first 4 terms, in ascending powers of x, of the expansion of $(1 - 3x)^7$. [3]

(ii) Find the coefficient of x^3 in the expansion of $(5 + 2x)(1 - 3x)^7$. [2]

(b) Find the term which is independent of x in the expansion of $\left(x^2 + \frac{2}{x}\right)^9$. [3]

9 **(i)** Given that $y = \frac{x + 2}{(4x + 12)^{\frac{1}{2}}}$, show that $\frac{dy}{dx} = \frac{k(x + 4)}{(4x + 12)^{\frac{3}{2}}}$, where k is a constant to be found. [5]

(ii) Hence evaluate $\int_1^{13} \frac{x + 4}{(4x + 12)^{\frac{3}{2}}}\,dx$. [3]

10 **(a)** Given that $\log_p X = 6$ and $\log_p Y = 4$, find the value of

(i) $\log_p \left(\frac{X^2}{Y}\right)$, [2]

(ii) $\log_Y X$. [2]

(b) Find the value of 2^z, where $z = 5 + \log_2 3$. [3]

(c) Express $\sqrt{512}$ as a power of 4. [2]

11 **(a)** Solve, for $0 < x < 3$ radians, the equation $4\sin x - 3 = 0$, giving your answers correct to 2 decimal places. [3]

(b) Solve, for $0° < y < 360°$, the equation $4\operatorname{cosec} y = 6\sin y + \cot y$. [6]

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12 Answer only **one** of the following two alternatives.

EITHER

It is given that $\mathrm{f}(x) = 4x^2 + kx + k$.

(i) Find the set of values of k for which the equation $\mathrm{f}(x) = 3$ has no real roots. [5]

In the case where $k = 10$,

(ii) express $\mathrm{f}(x)$ in the form $(ax + b)^2 + c$, [3]

(iii) find the least value of $\mathrm{f}(x)$ and the value of x for which this least value occurs. [2]

OR

The functions f, g and h are defined, for $x \in \mathbb{R}$, by

$$\mathrm{f}(x) = x^2 + 1,$$

$$\mathrm{g}(x) = 2x - 5,$$

$$\mathrm{h}(x) = 2^x.$$

(i) Write down the range of f. [1]

(ii) Find the value of gf(3). [2]

(iii) Solve the equation $\mathrm{fg}(x) = \mathrm{g}^{-1}(15)$. [5]

(iv) On the same axes, sketch the graph of $y = \mathrm{h}(x)$ and the graph of the inverse function $y = \mathrm{h}^{-1}(x)$, indicating clearly which graph represents h and which graph represents h^{-1}. [2]

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